Zachary Deutsch

zachary.deutsch@duke.edu | (312) 989-5658 | zacharydeutsch.com | LinkedIn

Education

Duke University — Durham, NC
B.S. Mathematics, B.S. Statistics, B.S.E. Mechanical Engineering, Minor in Philosophy, GPA: 3.7 — August 2024 – May 2028
Coursework: Linear Algebra, Differential Equations, Probability, Stochastic Finance, Data Structures & Algorithms, Statistics

University of Oxford — Oxford, England
Study Abroad: Philosophy & Ethics of AI — June – August 2025

Experience

Sonos Inc. — Boston, MA
Data Engineering Intern — June – August 2026
- Built dbt models in Snowflake integrating telemetry from 55M products with purchase and app activity data to score 16M customers weekly for Sonos Membership.

Veda Research — (Remote) New York, NY
Quantitative Research Intern — June 2026 – Present
- Built and tested 16 BTCUSDT trading strategies using 229M market events and 150 features, with walk-forward backtesting

Duke University, Oca Lab — Durham, NC
AI Research Analyst — October 2025 – Present
- Benchmarked LLM accuracy across 4,600 physics-question trials and analyzed responses from 38 subjects to assess error detection and confidence.

Duke University, Hickey Lab — Durham, NC
Machine Learning Researcher — November 2024 – Present
- Built and evaluated a cell-type annotation pipeline combining 4 clustering methods and 4 LLMs on 220K spatial-proteomic cells.

Tel Aviv University, Center for AI and Data Science — Tel Aviv, Israel
Machine Learning Research Intern — May – August 2025
- Compared quantile and cosine features for hand and wrist movement classification from muscle signals, achieving 92.3% test accuracy with subject-specific ExtraTrees models across 20 participants.

Publications

- **Deutsch, Z.**, Staupe, R., & Bartoszewicz, M. G. (forthcoming). *AI as a Catastrophic Imaginary.* Routledge Handbook of Catastrophic Imaginaries.
- **Deutsch, Z.**, Surguladze, N., Miao, Y., & Hickey, J. W. (under review). *Evaluation of Clustering Methods and Large Language Models for CODEX Cell Type Annotation.* EXO Beyond the Cell.
- **Deutsch, Z.**, Eva, R., & Oca, S. R. (2026). *Virtue Ethics and Student Use of Large Language Models in Engineering Education.* IEEE Frontiers in Education Conference.
- **Deutsch, Z.**, & Oca, S. R. (2026). *Evaluating Engineering Students' Detection of Hallucinations in Large Language Model Solutions.* American Society for Engineering Education Annual Conference.
- **Deutsch, Z.** (2026). *Superintelligence, Instrumental Convergence, and the Limits of AI Apocalypse.* AI and Ethics.

Skills, Awards, & Interests

Technical: Python, SQL | NumPy, pandas, scikit-learn | Git
Languages: Japanese (Limited Working Proficiency), Spanish (Limited Working Proficiency)
Awards: 1st Place J.P. Morgan GenerationTech, Google Scholar, 1st Place Bass Connections Fortin Foundation Showcase
Fellowships: Kenan Institute of Ethics Summer Fellow ($5k), Transformative Ideas Fellow ($1k), Character Forward Fellow ($1k)
Service: Technical Program Committee Member, IEEE Frontiers in Education Conference, 2026
Interests: Espresso, Liszt, Stoicism, Nietzsche, Cycling